

AQI MODULE — TECHNICAL DOCUMENTATION

1. AQI MODULE OVERVIEW

1.1 Section Overview

This module processes and analyzes **air quality data across Canadian monitoring stations**, converting raw hourly pollutant measurements into standardized **Air Quality Index (AQI)** metrics.

The system integrates:

- Hourly pollutant concentration data (NAPS)
- Station metadata (NAPS Station file)
- Time dimension (Model Calendar)
- Derived AQI metrics (daily + aggregated)

Primary Objective

To transform raw environmental sensor data into:

- Daily AQI values
- Station-level AQI indicators
- Dominant pollutant identification
- BI-ready datasets for analysis and visualization

2. DATA SOURCES & FILE STRUCTURE

2.1 Raw Data Sources

NAPS Air Monitoring Data

- Source: National Air Pollution Surveillance (NAPS)
- Structure: Hourly pollutant readings

From your pipeline:

```
AllPollutantsHourly/  
CO_2010.csv  
NO2_2011.csv  
O3_2024.csv
```

Combined AQI Files (Processed)

From your intermediate outputs:

AQI_CO_2010_2024.csv
AQI_NO2_2010_2024.csv
AQI_O3_2010_2024.csv
...

 Final Output (Your Clean Layer)

From your screenshot:

AQI_2015_2024_FINAL.parquet
AQI_2015_2024_new.parquet

 Station Metadata

From NAPS station file:

Column	Description
NAPS_ID	Unique station ID
Station_Name	Station name
City	Location
Province	Province
Latitude	Coordinate
Longitude	Coordinate

3. RAW DATA STRUCTURE (HOURLY FORMAT)

3.1 Input Dataset Schema

Each raw file contains:

- Station-level measurements
- 24 hourly columns

Example:

Column	Description
Pollutant	Type (O3, CO, PM2.5, etc.)
NAPS ID	Station ID
Date	Observation date

Column	Description
H01–H24	Hourly concentration values

3.2 Special Data Characteristics

✓ Wide-format structure

- Data initially stored as:

H01, H02, H03 ... H24

✓ Invalid value encoding

-999 → No observation (treated as NULL)

✓ Mixed pollutants per file

Handled by:

iterate per pollutant + per year

4. DATA TRANSFORMATION PIPELINE

4.1 Step 1 — Reshape Hourly Data

Transformation:

Wide format → Long format

Process:

```
melt(
  id_vars=[station, date, pollutant],
  value_vars=H01–H24
)
```

Result:

Datetime	Concentration
2024-01-01 01:00	value

4.2 Step 2 — Data Cleaning

✓ Replace invalid values:

-999 → NaN

✓ Remove unwanted pollutants:

Exclude:

NO

NOX

(Observed in your notebook logic)

4.3 Step 3 — Unit Conversion

✓ Ozone (O₃)

ppb → ppm

Conversion:

value / 1000

● Important

Required because:

- AQI breakpoints defined in ppm (EPA standard)

4.4 Step 4 — Daily Metric Calculation (CRITICAL)

Different pollutants use different aggregation rules:

✓ O₃ & CO

Daily metric = MAX(8-hour rolling average)

✓ NO₂ & SO₂

Daily metric = MAX(hourly value)

✓ PM_{2.5} & PM₁₀

Daily metric = MEAN(24-hour values)

✓ **Result:**

| Date | Pollutant | C (Daily Concentration) |

5. AQI COMPUTATION MODEL

5.1 EPA Breakpoint System

Each pollutant uses piecewise ranges:

Example (O₃):

Range (ppm)	AQI
0.000–0.054	0–50
0.055–0.070	51–100
...	...

5.2 Truncation Rule (IMPORTANT)

Before AQI calculation:

O₃ → 3 decimals

PM_{2.5} → 1 decimal

PM₁₀ → integer

CO → 1 decimal

NO₂, SO₂ → integer

(No rounding — truncation only)

5.3 AQI Formula

AQI =
(AQI_hi – AQI_lo)

(C_hi – C_lo)

× (C – C_lo)

+ AQI_lo

✓ **Output:**

C	AQI
Concentration	Index

✓ 5.4 Mathematical Representation (Add this)

For pollutant p at station s on day t:

$C(p,s,t)$ = aggregated concentration (from hourly data)

$AQI(p,s,t) =$
 $(AQI_hi - AQI_lo) / (C_hi - C_lo) \times (C - C_lo) + AQI_lo$

✓ Station AQI:

$AQI_station(s,t) = \max (AQI(p,s,t)) \ \forall \text{ pollutants } p$

✓ Dominant pollutant:

$Dominant_Pollutant(s,t) = \operatorname{argmax} (AQI(p,s,t))$

6. STATION-LEVEL AQI DERIVATION

6.1 Daily Station AQI

For each station and day:

$Station_AQI = \text{MAX}(AQI \text{ across pollutants})$

6.2 Dominant Pollutant

$Dominant_Pollutant =$
 pollutant with highest AQI

✓ Output Table (Your Power BI Fact Table)

From your screenshot:

Column	Description
NAPS_ID	Station ID

Column	Description
Date	Date
Pollutant	Pollutant type
Pollutant_Conc	Daily value
Pollutant_AQI	AQI per pollutant
Station_AQI	Max AQI
Dominant_Pollutant	Driver

7. FINAL DATASET CHARACTERISTICS

7.1 Grain

NAPS_ID × Date × Pollutant

7.2 Data Types

Column	Type
AQI	float32
Concentration	float32
Pollutant	category
NAPS_ID	int32

✓ Optimization

- Float64 → float32
- Object → category
- Saved as Parquet

8. DATA VALIDATION

✓ Uniqueness Check

No duplicates:
(NAPS_ID + Date + Pollutant)

✓ Consistency Check

Station_AQI = MAX(Pollutant_AQI)

✓ Missing Data Handling

- Rows with NULL AQI removed
- Invalid readings excluded

9. BI MODEL INTEGRATION

9.1 Fact Table

✓ Fact_AQI

From your Power BI view:

Column	Purpose
Pollutant_AQI	pollutant-level metric
Station_AQI	station-level metric
Dominant_Pollutant	key analysis field

Fact_AQI serves as the **central analytical fact table**, enabling:

- Drill-down from AQI → pollutant level
- Aggregation from station → province → region
- Filtering by pollutant, date, and AQI category
- Calculation of KPIs such as:
 - % Unhealthy Days
 - Dominant Pollutant Frequency
 - Average AQI trends

9.2 Dimension Tables Used

✓ Dim_Station_NAPS

- Joins on NAPS_ID

✓ Model Calendar

- Joins on Date

9.3 AQI Legend Table

From your screenshot:

Category	Range
Good	0–50
Moderate	51–100
Unhealthy	101–150
Very Unhealthy	201–300

Used for:

- color mapping
- report visuals

✓ 9.4 Table Grain Definitions

Table	Grain
Fact_AQI	NAPS_ID × Date × Pollutant
Station AQI	NAPS_ID × Date
Dim_Station_NAPS	NAPS_ID
Wildfire	Year × Province
Correlation	City × Month × Pollutant

10. EXTENDED ANALYSIS (FROM MODEL)

✓ Climate–AQI Correlation Table

From your screenshot:

Field	Description
Weather_Variable	climate factor
Pollutant	pollutant
Correlation	statistical result

✓ Wildfire Integration

From your Power BI screenshot:

Column	Description
Year	Year
Province	Region
Fires	Count
Total Area (HA)	burned area
Max Area	largest fire

✓ Role

- Supports AQI variation analysis
- Links pollution spikes to fire activity

✓ 11. MODEL ASSUMPTIONS & LIMITATIONS

✓ Assumptions

- AQI breakpoints follow **EPA standards**
- Hourly data sufficiently represents daily pollution behavior
- Rolling averages capture exposure correctly
- Missing values (NaN) represent no valid observation

✓ Limitations

- AQI derived from US-EPA model, not Canadian AQHI
- Station coverage is uneven across regions
- Pollutant mix may not fully represent air toxicity
- Extreme values may be influenced by local factors (fires, industry)